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GSR sensor output value internal calibration

The Shimmer GSR:

For the Shimmer GSR, please proceed to the article attached below ('*Shimmer GSR User Guide', P8-9*) for a detailed description on conductance computation.

The full scale range of the sensor can be set to a number of different preset values to allow accurate results across the full measurement range. Table 1 outlines the different settings and corresponding full scale ranges.

Setting	Full Scale Range	
	Resistance	Conductance
0	10 kΩ to 56 kΩ	100 µS to 17.9 µS
1	56 kΩ to 220 kΩ	17.9 µS to 4.5 µS
2	220 kΩ to 680 kΩ	4.5 µS to 1.5 µS
3	680 kΩ to 4.7 MΩ	1.5 µS to 0.2 µS
4	Auto-Range	

Table 1: Range of legitimate setting values for the GSR signal Range

The Shimmer hardware contains an internal resistor network which works as a potential divider and provides a voltage that can be converted by the Shimmer3 ADC to a 12-bit number that represents the external skin conductance. If you are using iMotions software for your GSR data acquisition, you will need to calculate the skin resistance from the ADC values. Each full scale range setting is mapped to a linear function which is used to calculate the skin conductance value:

$$y = p1*x + p2$$

where x = ADC output, y = Skin Conductance measured in µS. Table 2 below outlines the values of the linear function coefficients used by iMotions software, p1 and p2, for the different full scale range settings.

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Setting	Full Scale Range		Linear coefficients
	Resistance	**Conductance**	
0	10 kΩ to 56 kΩ	100 µS to 17.9 µS	p1 = 0.0373 p2 = -24.9915
1	56 kΩ to 220 kΩ	17.9 µS to 4.5 µS	p1 = 0.0054 p2 = -3.5194
2	220 kΩ to 680 kΩ	4.5 µS to 1.5 µS	p1 = 0.0015 p2 = -1.0163
3	680 kΩ to 4.7 MΩ	1.5 µS to 0.2 µS	p1 = 4.558e-04 p2 = -0.3014

Table 2: Coefficients for ADC value to skin conductance conversion

To further verify the GSR resistance values, please proceed to the article attached below (*'Shimmer GSR User Guide', P18*).

BIOPAC EDA / GSR 100C:

For BIOPAC EDA / GSR 100C, 'EDA100C CALIBRATION' notes that the EDA100C has scale setting in units of “µS per volt.” This is identical to the respective scale setting of µmho or micromho per volt. Namely, 1 µS = 1 µmho.

Setup calibration instructions are given in the articles attached below (*'Electrodermal Response (EDR) with an MP System' P4, 'EDA100C product Sheet'*):

- Lower frequency response at DC:

In the scaling window, set the input voltages so they map to the DC conductance ranges indicated by the sensitivity setting. For example, if the EDA100C is set to a Gain of 5 µS/V, then 0 V will map to 0 µS or infinite resistance, and 1 V will map to 5 µS or 200 kohm.

- Lower frequency response at 0.05 Hz:

In the scaling window, set the input voltages so they map to the “0.05 Hz” conductance ranges indicated by the sensitivity setting. For example if the EDA100C is set to a Gain of 5 µS/V, then 0 V will map to X µSs and 1 V will map to (X+5) µS. Where “X” is the mean conductance being recorded.

- To verify the Gain setting of the EDA100C:

1. Calibrate AcqKnowledge as detailed above for lower frequency response at DC.

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2. Place the lower frequency response to DC.
3. Set the Gain switch on the EDA100C to 5μS/V.
4. Perform measurement with electrodes disconnected. AcqKnowledge should produce a reading of 0 μS.
5. Insulate a 100 kohm resistor and place it from electrode pad to electrode pad (resistor must be insulated from fingers).
6. Perform measurement with electrode-resistor setup. AcqKnowledge should produce a reading of 10 μS.

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